Unit 2 Quiz - Solutions

Knowledge

Part A: Multiple Choice .Write the CAPITAL LETTER corresponding to the correct answer on the line provided.

1. If $f(x)=\sqrt{x^2-1}$ and, $g(x)=x+1$ which expression is equal to $[g(f(x))]'$? **C**

   (A) $1+\sqrt{x^2-1}$ (B) $\frac{x+1}{\sqrt{x^2+2x}}$ (C) $\frac{x}{\sqrt{x^2-1}}$ (D) $\sqrt{x^2+2x}$

2. For what value(s) of x do the tangents to the curves $f(x)=\left(x^2+1\right)^3$ and $g(x)=3x^2$ have the same slope? **D**

   (A) $x=-\sqrt{2},\sqrt{2}$ (B) $x=-1,0,1$ (C) $x=0,\pm\sqrt{2}$ (D) $x=0$

3. The slope of tangent line to the curve $f(x)=\frac{4}{\sqrt[3]{x}}$ at point x=8 is: **C**

   (A) $-\frac{1}{48}$ (B) $\frac{1}{48}$ (C) $-\frac{1}{12}$ (D) $\frac{1}{12}$

4. Find $f'(-1)$ if $f(x)=ax+\frac{b}{x}$, **B**

   (A) $a+b$ (B) $a-b$ (C) $a+2b$ (D) $-a+b$

Part B: Full Solution

5. Differentiate the following. Leave answers in simplified form with positive exponents only.

a) $f(x)=\left(1+\sqrt[4]{x^3}\right)\left(1+x^2\right)^2$ ✓✓✓

$$f(x)=\left(1+x^{\frac{3}{4}}\right)\left(1+2x^2+x^4\right)$$

$$f(x)=1+2x^2+x^4+x^{\frac{3}{4}}+2x^{\frac{11}{4}}+x^{\frac{19}{4}}$$

$$f'(x)=4x+4x^3+\frac{3}{4}x^{\frac{-1}{4}}+\frac{11}{2}x^{\frac{7}{4}}+\frac{19}{4}x^{\frac{15}{4}}$$

$$f'(x)=4x+4x^3+\frac{3}{4x^{\frac{1}{4}}}+\frac{11}{2}x^{\frac{7}{4}}+\frac{19}{4}x^{\frac{15}{4}}$$

b) $f(x)=\dfrac{2x^2+6\sqrt[5]{x^2}+\pi x}{\sqrt[3]{x}}$ ✓✓✓✓

$$f(x)=\frac{2x^2}{x^{\frac{1}{3}}}+\frac{6x^{\frac{2}{5}}}{x^{\frac{1}{3}}}+\frac{\pi x}{x^{\frac{1}{3}}}$$

$$f(x)=2x^{\frac{5}{3}}+6x^{\frac{1}{15}}+\pi x^{\frac{2}{3}}$$

$$f'(x)=\frac{10}{3}x^{\frac{2}{3}}+\frac{6}{15}x^{\frac{-14}{15}}+\frac{2\pi}{3}x^{\frac{-1}{3}}$$

$$f'(x)=\frac{10}{3}x^{\frac{2}{3}}+\frac{6}{15x^{\frac{14}{15}}}+\frac{2\pi}{3x^{\frac{1}{3}}}$$

c) $y=\dfrac{(3x^2-5)^3}{\sqrt[3]{x+1}}$ ✓✓✓✓

$$y=\left(3x^2-5\right)^3(x+1)^{-\frac{1}{3}}$$

$$y'=3\left(3x^2-5\right)^2(6x)(x+1)^{-\frac{1}{3}}-\frac{1}{3}(x+1)^{-\frac{4}{3}}\left(3x^2-5\right)^3$$

$$=\frac{1}{3}\left(3x^2-5\right)^2(x+1)^{-\frac{4}{3}}\left[54x(x+1)-\left(3x^2-5\right)\right]$$

$$=\frac{1}{3}\left(3x^2-5\right)^2(x+1)^{-\frac{4}{3}}\left(51x^2+54x+5\right)$$

$$=\frac{\left(3x^2-5\right)^2\left(51x^2+54x+5\right)}{3(x+1)^{\frac{4}{3}}}$$

d) $y=x+\sqrt{x^2+\sqrt{x^2-x}}$ ✓✓✓✓

$$y'=1+\frac{2x+\dfrac{2x-1}{2\sqrt{x^2-x}}}{2\sqrt{x^2+\sqrt{x^2-x}}}$$

$$=1+\frac{4x\sqrt{x^2-x}+2x-1}{4\left(\sqrt{x^2-x}\right)\sqrt{x^2+\sqrt{x^2-x}}}$$

6. Given $y=u^3+\frac{1}{2}u^2$ and $\frac{dy}{dx}=-20$ when $u=1$, find the value of $\frac{du}{dx}$.

$$\frac{dy}{dx}=\frac{dy}{du}\times\frac{du}{dx}$$
$$\left.\frac{dy}{dx}\right|_{u=1}=\left.\left(3u^2+u\right)\right|_{u=1}\times\left.\frac{du}{dx}\right|_{u=1}$$
$$-20=4\left.\frac{du}{dx}\right|_{u=1}$$
$$\left.\frac{du}{dx}\right|_{u=1}=-5$$

Application

1. If $f(x)=\sqrt[3]{x^2}+\frac{4}{(x-2)^2}$, find $f''(1)$.

$$f(x)=x^{\frac{2}{3}}+4(x-2)^{-2}$$
$$f'(x)=\frac{2}{3}x^{\frac{-1}{3}}-8(x-2)^{-3}$$
$$f''(x)=\frac{-2}{9}x^{\frac{-4}{3}}+24(x-2)^{-4}$$
$$f''(1)=\frac{-2}{9}+24$$
$$\boxed{f''(1)=\frac{214}{9}}$$

2. Given $f'(1)=4, g'(1)=-2, f(1)=1$, and $g(1)=1$ find $h'(1)$ if $h(x)=x\,g\left(f(x^2)\right)$.

$$h'(x)=g\left(f(x^2)\right)+x(2x)f'(x^2)g'\left(f(x^2)\right)$$
$$h'(1)=g\left(f(1)\right)+(2)f'(1)g'\left(f(1)\right)$$
$$h'(1)=g(1)+(2)(4)g'(1)$$
$$h'(1)=1+(8)(-2)$$
$$\boxed{h'(1)=-15}$$

3. Consider the graph of $f(x)=x^3-9x$. The tangent has a y-intercept of 16. Determine the exact area of the triangle formed in the second quadrant by the tangent line and the axes.

Let $A\left(a,a^3-9a\right)$ represent the tangency point.

since tangent line passes through point $B(0,16)$, we have:

$$m_{AB}=\frac{a^3-9a-16}{a-0}$$
$$m_T=f'(a)=3a^2-9$$
$$\frac{a^3-9a-16}{a}=3a^2-9$$
$$a^3-9a-16=3a^3-9a$$
$$-16=2a^3$$
$$-8=a^3$$
$$a=-2\rightarrow m_T=3(-2)^2-9$$
$$m_T=3$$

Equation of tangent line: $y=3x+16$

$$x\text{-int}: x=\frac{-16}{3}$$
$$A_\Delta=\frac{1}{2}(16)\left(\frac{16}{3}\right)$$
$$=\frac{128}{3}\text{ units}^2$$

4. A particle moves according to the equation $s(t)=4\left(\frac{4}{3}-\frac{1}{35}t^2\right)t^{\frac{3}{2}}$ where t is in seconds and $s(t)$ is the distance in metres. Determine the following:

a) the velocity of a particle when its acceleration is zero

$$s(t)=\frac{16}{3}t^{\frac{3}{2}}-\frac{4}{35}t^{\frac{7}{2}}$$

$$v(t)=s'(t)=8t^{\frac{1}{2}}-\frac{2}{5}t^{\frac{5}{2}}$$

$$a(t)=s''(t)=4t^{\frac{-1}{2}}-t^{\frac{3}{2}}$$

$$a(t)=0\Rightarrow t^{\frac{-1}{2}}\left(4-t^2\right)=0$$

$$4-t^2=0\rightarrow t^2=4\rightarrow \boxed{t=2}$$

$$v(2)=8(2)^{\frac{1}{2}}-\frac{2}{5}(2)^{\frac{5}{2}}$$

$$v(2)=\frac{32}{5}\sqrt{2}\approx 9.1\text{ m/s}$$

b) exact time(s) at which the velocity is equal to zero.

$$v(t)=0\Rightarrow 8t^{\frac{1}{2}}-\frac{2}{5}t^{\frac{5}{2}}=0$$

$$t^{\frac{1}{2}}\left(8-\frac{2}{5}t^2\right)=0$$

$$t^2=20\rightarrow \boxed{t=2\sqrt{5}\text{ s}}\quad\text{or}\quad\boxed{t=0}$$

Thinking

1. Consider the rational function $f(x)=\frac{ax-b}{bx+1}$, where **a** and **b** are constants. Determine values for **a** and **b** such that the tangent lines to the graph of f(x) at points $x=0$ and $x=-\frac{2}{3}$ are parallel to the line $y=5x-2$.✓✓✓✓✓

$$\frac{dy}{dx}=\frac{a+b^2}{(bx+1)^2}$$

$$\left.\frac{dy}{dx}\right|_{x=0}=5\rightarrow a+b^2=5 \qquad (1)$$

$$\left.\frac{dy}{dx}\right|_{x=\frac{-2}{3}}=5\rightarrow a+b^2=5\left(\frac{-2}{3}b+1\right)^2 \qquad (2)$$

$$\text{Sub. (1) into (2): } \cancel{5}=\cancel{5}\left(\frac{-2}{3}b+1\right)^2\rightarrow \frac{-2}{3}b+1=\pm 1$$

$$b=0 \text{ or } \boxed{b=3}\xrightarrow{\text{Sub. into (1)}}\boxed{a=2}$$

b = 0 Inadmissible

(makes the function a linear function)